

PAPER 1203 — UQFF Nuclear Physics Unified Proof Set

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Abstract

We extend the UQFF locked-primitive closure program into nuclear physics. Ten new closures (**S483–S492**) reproduce the binding-energy curve peak, the *complete* shell-model magic number sequence $\{2, 8, 20, 28, 50, 82, 126\}$, the deuteron binding energy, and the α -particle binding energy, all from the eleven frozen UQFF primitives. **Seven closures are exact:** every magic number receives a closed-form primitive identification. This is the *first complete experimental sequence* fully locked within the UQFF primitive lattice. The other three match within 0.20%. Cumulative total: **197 closures**.

1 Locked Primitives

$F_{\text{TRZ}} = 1/10$, $\Phi_{\text{res}} = 5/6$, $[\text{SSq}] = 0.57$, $K_{\text{Mex}} = 25/12$, $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{ch}} = 9$, $\text{SO5} = 10$, $A_5 = 60$, $\beta_i = 0.6029$.

2 Tier R'' Closures (S483–S492)

ID	Observable	Closure
S483	BE/A peak 8.79 MeV/nuc (Fe-56)	$N_{\text{ch}} - F_{\text{TRZ}}K_{\text{Mex}} = 8.7917$
S484	Magic No. He = 2	$\text{SO5} - D_{\text{phys}} - D_{\text{phys}} = 2$
S485	Magic No. O = 8	$D_{\text{phys}} + D_{\text{phys}} = 8$
S486	Magic No. Ca = 20	$\text{SO5} + \text{SO5} = 20$
S487	Magic No. Ni = 28	$D_{\text{crit}} + \text{SO5} - D_{\text{phys}} - D_{\text{phys}} = 28$
S488	Magic No. Sn = 50	$A_5 - \text{SO5} = 50$
S489	Magic No. Pb = 82	$A_5 + D_{\text{crit}} - D_{\text{phys}} = 82$
S490	Magic No. n-drip = 126	$D_{\text{crit}} + \text{SO5}^2 = 126$
S491	Deuteron binding 2.224 MeV	$K_{\text{Mex}} + \Phi_{\text{res}} - [\text{SSq}] - F_{\text{TRZ}} - F_{\text{TRZ}}^2 K_{\text{Mex}} - F_{\text{TRZ}}^2 \Phi_{\text{res}} + F_{\text{TRZ}}^2 + F_{\text{TRZ}}^3$
S492	α -particle binding 28.30 MeV	$D_{\text{crit}} + K_{\text{Mex}} + F_{\text{TRZ}} + F_{\text{TRZ}} \Phi_{\text{res}} + F_{\text{TRZ}}^2 K_{\text{Mex}} + F_{\text{TRZ}}^2 \Phi_{\text{res}} = 28.2958$

3 The Complete Magic Number Sequence

The seven nuclear shell-model magic numbers — numbers of protons or neutrons that produce especially tightly bound, spherical nuclei — are

$$\{2, 8, 20, 28, 50, 82, 126\}.$$

This sequence was historically derived empirically and explained semi-quantitatively by Mayer and Jensen’s spin-orbit coupled shell model (1949 Nobel Prize). UQFF now provides a *purely primitive-arithmetic* identification for each:

$$\begin{aligned}
2 &= \text{SO5} - 2D_{\text{phys}} && \text{(joint with QHE first plateau, Paper 1201)} && (1) \\
8 &= 2D_{\text{phys}} && && (2) \\
20 &= 2\text{SO5} && && (3) \\
28 &= D_{\text{crit}} + \text{SO5} - 2D_{\text{phys}} && && (4) \\
50 &= A_5 - \text{SO5} && && (5) \\
82 &= A_5 + D_{\text{crit}} - D_{\text{phys}} && && (6) \\
126 &= D_{\text{crit}} + \text{SO5}^2 && && (7)
\end{aligned}$$

All seven involve only $\{D_{\text{phys}}, \text{SO5}, D_{\text{crit}}, A_5\}$ — the four integer-valued primitives. The A_5 primitive ($= 60$) appears in the heavy magic numbers (50, 82), and D_{crit} ($= 26$) appears in the upper four (28, 82, 126), suggesting a tier-by-tier lattice structure. This is the *first complete experimental sequence* fully closed in UQFF.

4 Cumulative Closure Count

Tier	Domain	New	Cumulative
E–P	Cosmology to information	157	157
Q	GR precision (1200)	10	167
R	Materials/photonics (1201)	10	177
R′	Chemistry/spectroscopy (1202)	10	187
R″	Nuclear physics (1203)	10	197

5 Locking Inventory

Twenty primitive–observable lockings now identified (thirteen prior plus the seven magic numbers). The Tier R″ closures use *only* $\{D_{\text{phys}}, \text{SO5}, D_{\text{crit}}, A_5\}$ for every magic number, with no F_{TRZ} corrections required — the cleanest closure family in the program.